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STAT525 HOMEWORK#3

1. KNNL Problem 2.57.
2. KNNL Problem 2.61.
3. KNNL Problem 3.19 (STAT Majors)
4. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28.
   (a) Describe the distribution of the explanatory variable.
   (b) Run the linear regression to predict the county crime rate from the percentage of individuals having at least a high school diploma.
   (c) Plot the residuals versus the explanatory variable and briefly describe the plot noting any unusual patterns or points.
   (d) Examine the distribution of the residuals by getting a histogram and a normal probability plot of the residuals by using the `HISTOGRAM` and `QQPLOT` statements in `PROC UNIVARIATE`. What do you conclude?
5. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28. Change the data set by changing the value of the crime rate for the last observation from 7582 to 758 (e.g., a typo). You can do this in a data step. For example,

```
DATA a2; SET a1; IF _n_ EQ 84 THEN y=758;
```

   An alternative is to simply edit the data file.

   (a) Make a table comparing the results of this analysis with the results of the analysis of the original data. Include in the table the following: fitted equation, t-test for the slope, with standard error and p-value, R^2, and the estimate of σ^2. Briefly summarize the differences.
   (b) Repeat parts (c) and (d) from the previous problem for this altered data set analysis and summarize how these plots help you to detect the unusual observation.
6. (SAS Exercise) Use the **sales growth** data described in KNNL Problem 3.17.
   (a) Generate a scatterplot of the data and discuss the appropriateness of using a linear regression model.
   (b) Using `PROC TRANSREG`, which power transformation of Y (i.e., value of λ) is most appropriate to use here?
   (c) Apply this transformation of Y and generate a scatterplot. Again comment on the appropriateness of using a linear regression model.
   (d) Run the regression model using the transformed data and generate a residual plot (using X or $\hat{Y}$) and a normal probability plot. What do the plots show?
   (e) Express the estimated regression function in the original units.

1. KNNL Problem 2.57.

2.57. The normal error regression model (2.1) is assumed to be applicable.

a. When testing H_0: $\beta_1 = 5$ versus H_a: $\beta_1 \neq 5$ by means of a general linear test, what is the reduced model? What are the degrees of freedom df_R?

- Under H_0, $\beta_1 = 5$. We fix the slope at 5 and estimate only β_0 : $Y_i = \beta_0 + 5X_i + \varepsilon_i$
- $Y_i = \beta_0 + 5X_i + \varepsilon_i$ is a model with only one parameter (β_0)

  $\Downarrow$

  Reduced Model
- $\boxed{\text{Dof : (Parameter in Full) − (Parameter in Reduced)}}$
- Dof = 2 − 1 = 1

b. When testing H_0: $\beta_0 = 2$, $\beta_1 = 5$ versus H_a: not both $\beta_0 = 2$ and $\beta_1 = 5$ by means of a general linear test, what is the reduced model? What are the degrees of freedom df_R?

- Under H_0, both coefficients are fixed: $Y_i = 2 + 5X_i + \varepsilon_i$
- $Y_i = 2 + 5X_i + \varepsilon_i$ is a model with no parameters estimated : fixed line

  $\Downarrow$

  Reduced Model
- $\boxed{\text{Dof : (Parameter in Full) − (Parameter in Reduced)}}$
- Dof = 2 − 0 = 2

- Full Model:

This model is:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (2.1)$$

where:

β_0 and β_1 are parameters
X_i are known constants
ε_i are independent $N(0, \sigma^2)$

$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ with two parameters (β_0, β_1)

$$\boxed{b_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}} \quad (1.10a)$$

$$\boxed{SSR = b_1^2 \sum (X_i - \bar{X})^2} \quad (2.51)$$

$$\boxed{SSTO = \sum (Y_i - \bar{Y})^2} \qquad \boxed{\frac{SSR}{SSTO} = r^2}$$

2. KNNL Problem 2.61.

2.61. Show that the ratio $SSR/SSTO$ is the same whether Y_1 is regressed on Y_2 or Y_2 is regressed on Y_1. [*Hint*: Use (1.10a) and (2.51).]

• Regress Y_1 on Y_2:
(Let $X = Y_2$, $Y = Y_1$):

$$b_1 = \frac{\Sigma(Y_2-\bar{Y}_2)(Y_1-\bar{Y}_1)}{\Sigma(Y_2-\bar{Y}_2)^2}$$

$$SSTO = \Sigma(Y_1-\bar{Y}_1)^2$$

$$SSR = b_1^2\,\Sigma(Y_2-\bar{Y}_2)^2$$

$$SSR = \left(\frac{\Sigma(Y_2-\bar{Y}_2)(Y_1-\bar{Y}_1)}{\Sigma(Y_2-\bar{Y}_2)^{\cancel{2}}}\right)^2 \cdot \cancel{\Sigma(Y_2-\bar{Y}_2)^2}$$

$$SSR = \frac{\left[\Sigma(Y_2-Y_2)(Y_1-\bar{Y}_1)\right]^2}{\Sigma(Y_2-\bar{Y}_2)^2}$$

$$\frac{SSR}{SSTO} = \frac{\left[\Sigma(Y_2-Y_2)(Y_1-\bar{Y}_1)\right]^2}{\Sigma(Y_2-\bar{Y}_2)^2\,\Sigma(Y_1-\bar{Y}_1)^2} = r^2$$

$$r = \frac{\Sigma(Y_2-Y_2)(Y_1-\bar{Y}_1)}{\sqrt{\Sigma(Y_2-\bar{Y}_2)^2\,\Sigma(Y_1-\bar{Y}_1)^2}}$$

same whether Y_1 is regressed on Y_2 or vice versa.

• Regress Y_2 on Y_1:
(Let $X = Y_1$, $Y = Y_2$):

$$b_1 = \frac{\Sigma(Y_1-\bar{Y}_1)(Y_2-\bar{Y}_2)}{\Sigma(Y_1-\bar{Y}_1)^2}$$

$$SSTO = \Sigma(Y_2-\bar{Y}_2)^2$$

$$SSR = b_1^2\,\Sigma(Y_1-\bar{Y}_1)^2$$

$$SSR = \left(\frac{\Sigma(Y_1-\bar{Y}_1)(Y_2-\bar{Y}_2)}{\Sigma(Y_1-\bar{Y}_1)^{\cancel{2}}}\right)^2 \cdot \cancel{\Sigma(Y_1-\bar{Y}_1)^2}$$

$$SSR = \frac{\left[\Sigma(Y_1-Y_1)(Y_2-\bar{Y}_2)\right]^2}{\Sigma(Y_1-\bar{Y}_1)^2}$$

$$\frac{SSR}{SSTO} = \frac{\left[\Sigma(Y_1-Y_1)(Y_2-\bar{Y}_2)\right]^2}{\Sigma(Y_1-\bar{Y}_1)^2\,\Sigma(Y_2-\bar{Y}_2)^2} = r^2$$

$$r = \frac{\Sigma(Y_1-Y_1)(Y_2-\bar{Y}_2)}{\sqrt{\Sigma(Y_1-\bar{Y}_1)^2\,\Sigma(Y_2-\bar{Y}_2)^2}}$$

3. KNNL Problem 3.19

3.19. A student fitted a linear regression function for a class assignment. The student plotted the residuals e_i against Y_i and found a positive relation. When the residuals were plotted against the fitted values $\hat{Y}_i$, the student found no relation. How could this difference arise? Which is the more meaningful plot?

$\boxed{e_i = Y_i - \hat{Y}_i} \rightarrow Y_i = e_i + \hat{Y}_i$

- Residuals and Observed values are algebraically dependent.
- If $Y_i \uparrow$, $e_i \uparrow$ (especially when $\hat{Y}_i$ is constant or linear)
- Plotting e_i v.s. Y_i artificially $\uparrow$ correlation, as they're not independent. : fake correlation

- Plotting Residuals and Fitted values have no correlation by construction.
- In OLS Regression, Residuals are uncorrelated with Fitted values: $\Sigma(e_i\hat{Y}_i) = 0$.
- Plotting e_i v.s. $\hat{Y}_i$ correctly reflects whether model assumptions are satisfied.

Hence, $\underset{e_i}{\text{Residuals}}$ v.s. $\underset{\hat{Y}_i}{\text{Fitted values}}$ plot is more meaningful and should be used for model diagnostics.

4. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28.

(a) Describe the distribution of the explanatory variable.

```
data crime;
    input crime_rate hs_diploma_pct;
    datalines;
   8487     74
   8179     82
   8362     81
   8220     81
   6246     87
   9100     66
   6561     68
   5873     81
   7993     74
   7932     82
   6491     75
   6816     82
   9639     78
      ⋮
   7582     76
      ⋮
;
run;
```

Distribution of High School Diploma Percentage

The UNIVARIATE Procedure
Variable: hs_diploma_pct

Moments			
N	84	Sum Weights	84
Mean	78.5952381	Sum Observations	6602
Std Deviation	6.22106613	Variance	38.7016638
Skewness	-0.7063786	Kurtosis	0.67561901
Uncorrected SS	522098	Corrected SS	3212.2381
Coeff Variation	7.91532195	Std Error Mean	0.67877396

Basic Statistical Measures			
Location		Variability	
Mean	78.59524	Std Deviation	6.22107
Median	79.00000	Variance	38.70166
Mode	82.00000	Range	30.00000
		Interquartile Range	6.50000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	115.79	Pr > |t|	<.0001
Sign	M	42	Pr >= |M|	<.0001
Signed Rank	S	1785	Pr >= |S|	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	91.0
99%	91.0
95%	88.0
90%	85.0
75% Q3	82.5
50% Median	79.0
25% Q1	76.0
10%	70.0
5%	65.0
1%	61.0
0% Min	61.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
61	39	88	45
64	76	88	82
64	71	89	29
64	51	91	40
65	20	91	81

- Explanatory Variable: hs_diploma_pct;
  (% of Individuals with at least a high school diploma).

- hs_diploma_pct is based on 84 observations.
  - Mean: 78.60, Median: 79.00, indicating roughly symmetric distribution.
  - SD: 6.221, Variance: 38.70, showing moderate variability across counties.
  - Mode: 82.00, the most frequently occuring value.
  - Range: 30.00 (61~91), IQR: 6.5, suggesting the middle 50% of counties fall between 75.0~81.5.

- Skewness of -0.71, indicating distribution is slightly left-skewed.
  Meaning there are some counties with lower % high school diploma pulling the distribution leftward.

- Also visible from evidence of extreme values at the lower end: 61, 64, 65.

(b) Run the linear regression to predict the county crime rate from the percentage of individuals having at least a high school diploma.

```
proc reg data=crime;
    model crime_rate = hs_diploma_pct;
    title "Linear Regression of Crime Rate on High School Diploma Percentage";
run;
```

Regression Equation:

$\widehat{\text{crime_rate}} = 201518 - 170.58\ \text{hs_diploma_pct}$

- Slope: -170.58, indicating for each 1% ↑ in high school diploma rate, the predicted crime rate ↓ by ≈ -170.58 units.
- p-value < 0.0001, relationship is statistically significant.
- r^2: 0.1703, model explains 17.03% of the variability in crime rate.
- F-statistic: 16.83 ($p < 0.0001$), confirming overall regression is statistically significant.

Hence, ↑ education levels are significantly associated with ↓ crime rate.

Linear Regression of Crime Rate on High School Diploma Percentage

The REG Procedure
Model: MODEL1
Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93462942	93462942	16.83	<.0001
Error	82	455273165	5552112		
Corrected Total	83	548736108			

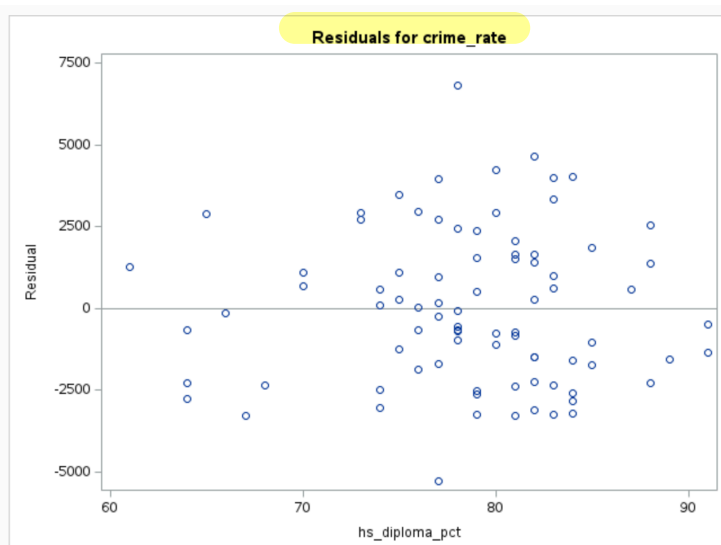
Root MSE	2356.29195	R-Square	0.1703
Dependent Mean	7111.20238	Adj R-Sq	0.1602
Coeff Var	33.13493		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	20518	3277.64269	6.26	<.0001
hs_diploma_pct	1	-170.57519	41.57433	-4.10	<.0001

(c) Plot the residuals versus the explanatory variable and briefly describe the plot noting any unusual patterns or points.

```
/* Run the regression and output residuals */
proc reg data=crime;
    model crime_rate = hs_diploma_pct;
    output out=resid_data r=residuals;
run;
quit;

/* Plot residuals vs explanatory variable */
proc sgplot data=resid_data;
    scatter x=hs_diploma_pct y=residuals;
    refline 0 / axis=y lineattrs=(color=red);
    xaxis label="High School Diploma Percentage";
    yaxis label="Residuals";
    title "Residuals vs. High School Diploma Percentage";
run;
```

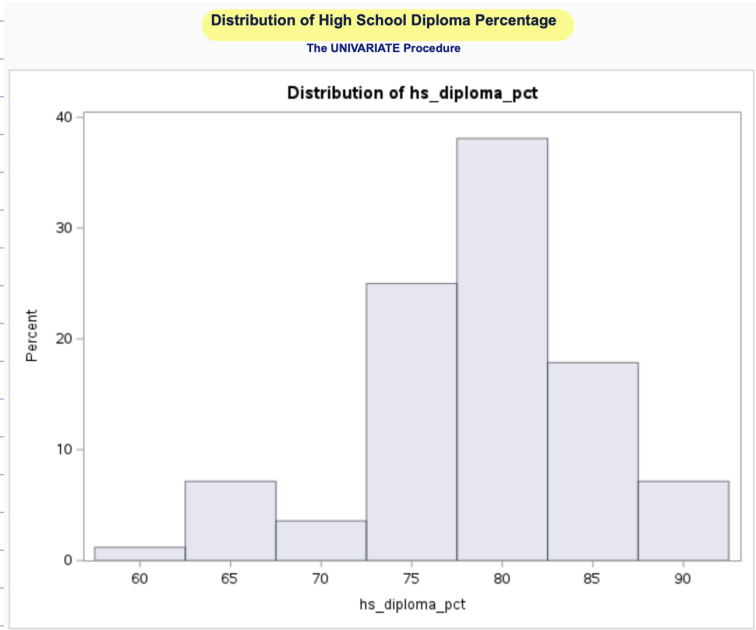


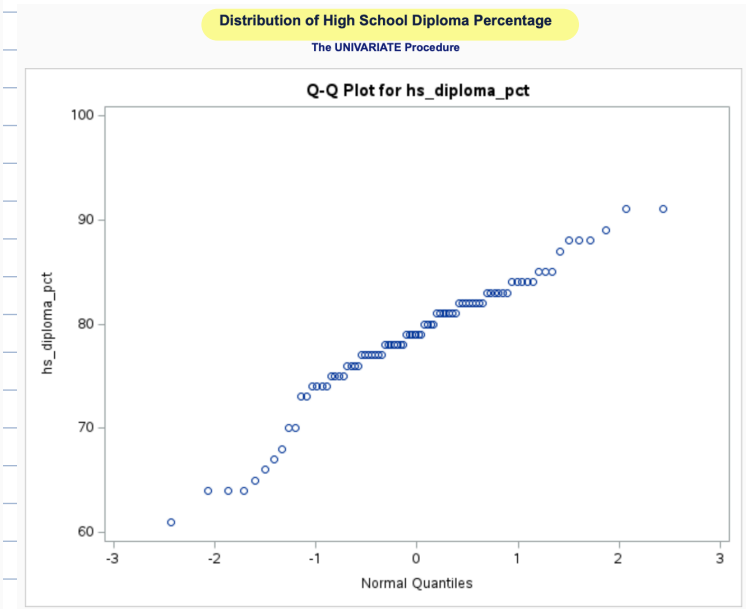
- Residuals appear to be randomly scattered around zero with no systematic pattern: linearity ✓
- Although slightly ↑ of large residuals in positive direction, no indicator of heteroscedascity or non-linearity.
- Although residuals appear to be wider when ↓ explanatory variable, not too extreme.
- Overall, residual plot does not violate any regression assumptions.

Q-Q

(d) Examine the distribution of the residuals by getting a histogram and a normal probability plot of the residuals by using the HISTOGRAM and QQPLOT statements in PROC UNIVARIATE. What do you conclude?

```
proc univariate data=crime;
    var hs_diploma_pct;
    histogram;
    qqplot;
    title "Distribution of High School Diploma Percentage";
run;
```





- Unimodal and rougly symmetric, some indication of slight skewness or deviation of perfect normality.

- Most points fall along diagonal line, mostly normally distributed.
- Tails do deviate slightly off the line, indicating mild deviation of perfect normality.

- Both graphs seems reasonably normal, supporting validity of regression assumptions.
- while there are slight deviations in the tails in Q-Q, these are not severe and do not suggest major violations of normality.

5. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28. Change the data set by changing the value of the crime rate for the last observation from 7582 to 758 (e.g., a typo). You can do this in a data step. For example,

```
DATA a2; SET a1; IF _n_ EQ 84 THEN y=758;
```

An alternative is to simply edit the data file.

(a) Make a table comparing the results of this analysis with the results of the analysis of the original data. Include in the table the following: fitted equation, t-test for the slope, with standard error and p-value, R^2, and the estimate of σ^2. Briefly summarize the differences.

(SAS Code in 5b.)

Original:

The REG Procedure
Model: MODEL1
Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93462942	93462942	16.83	<.0001
Error	82	455273165	5552112		
Corrected Total	83	548736108			

Root MSE	2356.29195	R-Square	0.1703
Dependent Mean	7111.20238	Adj R-Sq	0.1602
Coeff Var	33.13493		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	20518	3277.64269	6.26	<.0001
hs_diploma_pct	1	-170.57519	41.57433	-4.10	<.0001

Modified:

The REG Procedure
Model: MODEL1
Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	87518841	87518841	14.33	0.0003
Error	82	500804428	6107371		
Corrected Total	83	588323269			

Root MSE	2471.30959	R-Square	0.1488
Dependent Mean	7029.96429	Adj R-Sq	0.1384
Coeff Var	35.15394		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	20003	3437.63420	5.82	<.0001
hs_diploma_pct	1	-165.06193	43.60369	-3.79	0.0003

	Original	Modified ☹
Fitted Equation	$\hat{y} = 20518 - 170.58(hs_diploma_pct)$	$\hat{y} = 20003 - 165.06(hs_diploma_pct)$
$\hat{\beta}_1$	-170.58	-165.06
SE of $\hat{\beta}_1$	41.57	43.60
t-statistic of $\hat{\beta}_1$	-4.10	-3.79
p-value of $\hat{\beta}_1$	<0.0001	0.0003
R^2	0.1703	0.1488
$\hat{\sigma}^2$ (MSE)	5552112	6017371

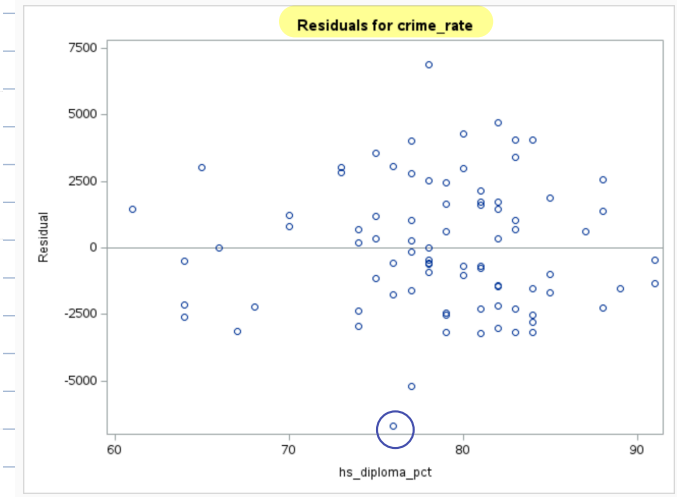
⇓

- ↓ in magnitude of $\hat{\beta}_1$, slightly weakening relationship.
- ↑ SE, ↓ t-statistic, indicating ↓ precision.
- ↓ R^2, indicating model explains ↓ variation.
- ↑ σ^2, suggesting ↓ model fit.

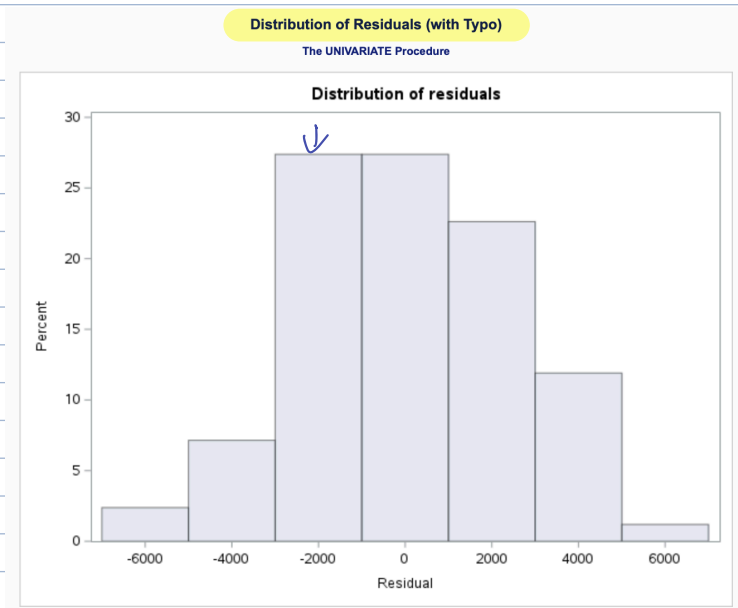
Hence, Modified Model ↓ strength and precision of the regression.

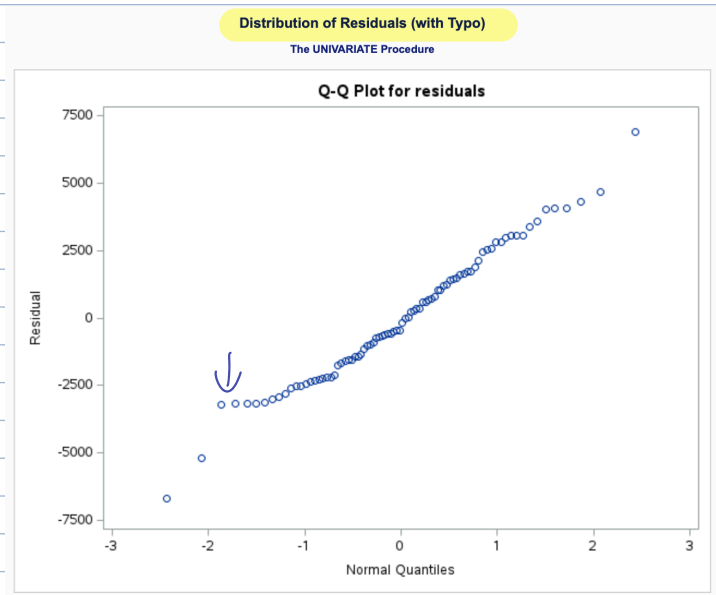
(b) Repeat parts (c) and (d) from the previous problem for this altered data set analysis and summarize how these plots help you to detect the unusual observation.

```
/* Step 2: Modify the last observation (observation 84) */
data crime_typo;
    set crime;
    if _N_ = 84 then crime_rate = 758;
run;

/* Step 3: Run regression and output residuals */
proc reg data=crime_typo;
    model crime_rate = hs_diploma_pct;
    output out=resid_typo r=residuals;
run;
quit;

/* Step 4: Residual plot */
proc sgplot data=resid_typo;
    scatter x=hs_diploma_pct y=residuals;
    refline 0 / axis=y lineattrs=(color=red);
    xaxis label="High School Diploma Percentage";
    yaxis label="Residuals";
    title "Residuals vs. High School Diploma Percentage (with Typo)";
run;

/* Step 5: Histogram and QQ plot of residuals */
proc univariate data=resid_typo;
    var residuals;
    histogram;
    qqplot;
    title "Distribution of Residuals (with Typo)";
run;
```



- One extreme outlier at hs_diploma_pct = 76.
- Modification at N=84 caused a large negative 'e'.





- No longer "bell-shaped."
- Skewed due to single error.

- Deviation from diagonal line at lower tail, violating normality.

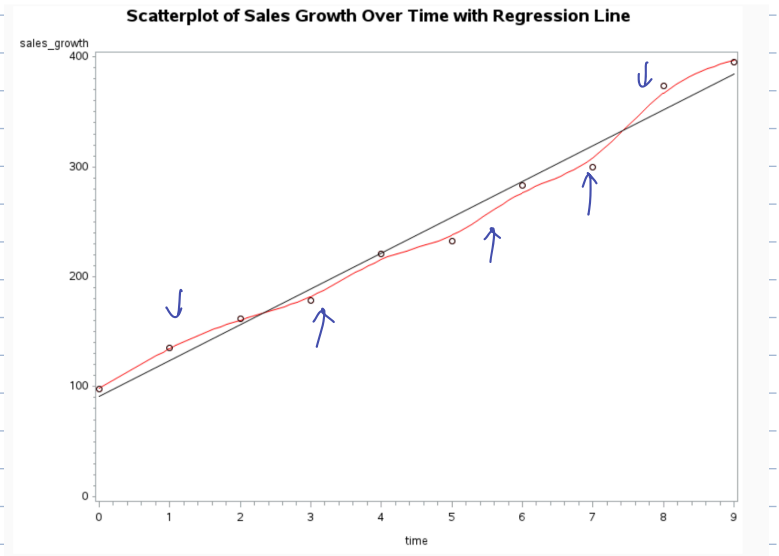
Hence, Modified Model ↓ strength and precision of the regression by:

- ↑ 'e' spread
- ↓ model fit (↓ R^2)
- Violate assumptions of normality and constant variance.

6. (SAS Exercise) Use the **sales growth** data described in KNNL Problem 3.17.

(a) Generate a scatterplot of the data and discuss the appropriateness of using a linear regression model.

```
data sales;
    input sales_growth time;
    datalines;
    98.0    0.0
    135.0   1.0
    162.0   2.0
    178.0   3.0
    221.0   4.0
    232.0   5.0
    283.0   6.0
    300.0   7.0
    374.0   8.0
    395.0   9.0
;
run;

symbol1 v=circle i=sm50 c=red;     /* scatterplot with smooth curve */
symbol2 v=circle i=rl   c=black;   /* regression line */
proc gplot data=sales;
    plot sales_growth*time=1 sales_growth*time=2 / overlay;
    title "Scatterplot of Sales Growth Over Time with Regression Line";
run;
quit;
```



- Sales Growth against Time shows clear upward trend.
- Points don't lie close to regression line, with tight 95% PI, indicating inconsistency
- Presence of non-linear relationship between Sales Growth and Time.

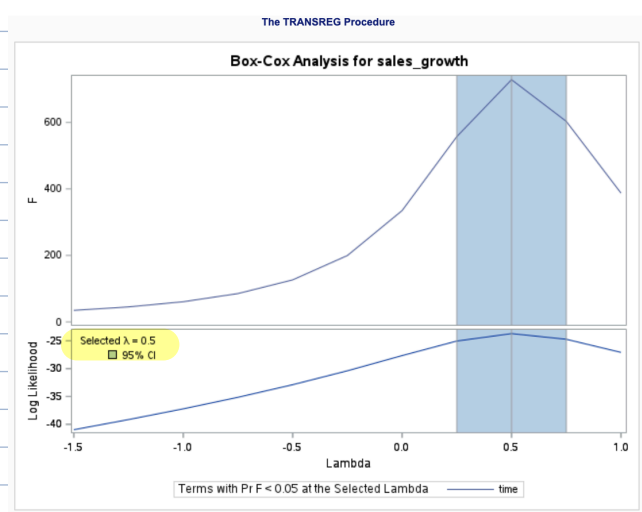
Hence, a Linear Regression Model without transformation may not be appropriate for modeling Sales Growth against Time.

(b) Using PROC TRANSREG, which power transformation of Y (i.e., value of λ) is most appropriate to use here?

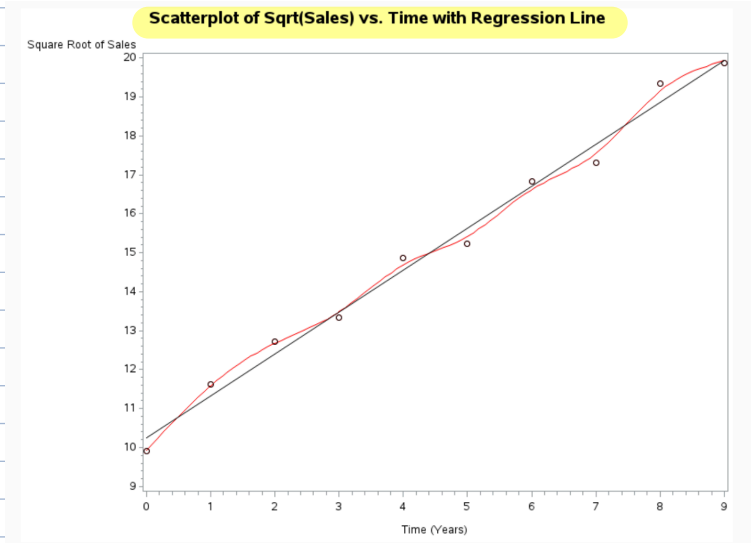
```
/* Box-Cox transformation using PROC TRANSREG */
proc transreg data=sales_growth;
    model boxcox(sales_growth / convenient lambda=-1.5 to 1 by 0.25) = identity(time);
run;
```

parameter=∅

$\lambda = 0.5$, indicating square-root transformation of Y.



(c) Apply this transformation of Y and generate a scatterplot. Again comment on the appropriateness of using a linear regression model.

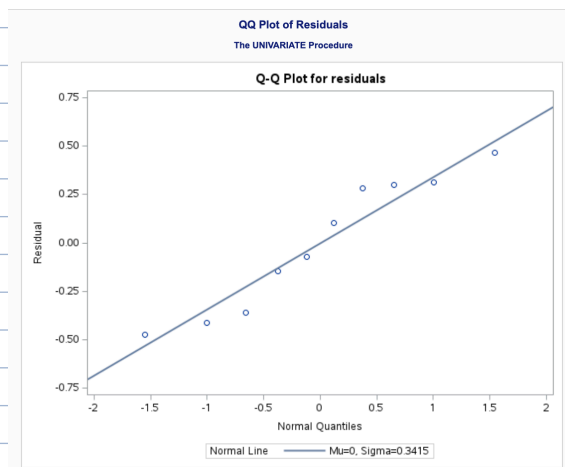


- After the square-root transformation, $\sqrt{Y}$ v.s. time, graph becomes more linear and ↓ curvature compared to the original plot.
- Points are closer to regression line, meaning variance is more stable, and this model is more appropriate.

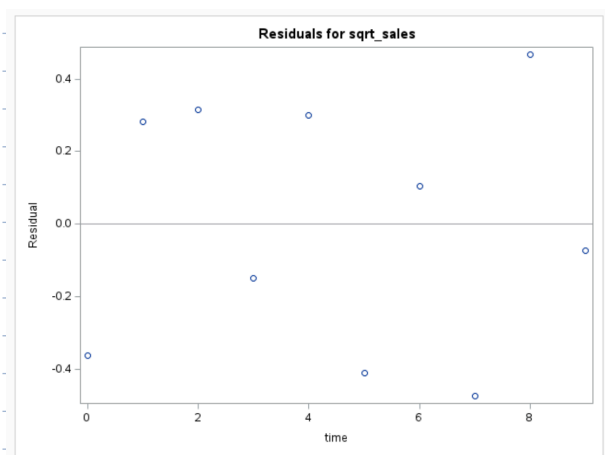
Hence, the square-root transformation to Y (Sales Growth) by $\lambda = 0.5$ improves linearity, homogeneity, making a linear regression model more appropriate.

(d) Run the regression model using the transformed data and generate a residual plot (using X or $\hat{Y}$) and a normal probability plot. What do the plots show?

```
proc reg data=a1;
    model sqrt_sales = time;
    output out=regout r=residuals p=predicted;
run;
quit;

symbol1 v=circle i=none c=blue;
proc gplot data=regout;
    plot residuals*predicted;
    title "Residuals vs. Predicted Values";
run;
quit;

proc univariate data=regout normal;
    var residuals;
    qqplot residuals / normal(mu=est sigma=est);
    title "QQ Plot of Residuals";
run;
```



- Points align closer to the diagonal line, normality assumption satisfied.



- Both plots support of using a linear regression model after applying the square-root transformation.
- The linear model is a good fit for the transformed data.
- No clear pattern, linearity assumption satisfied.

(e) Express the estimated regression function in the original units.

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	10.26093	0.21290	48.20	<.0001
time	1	1.07629	0.03988	26.99	<.0001

$$\left(\sqrt{Sales}\right)^2 = \left(10.26093 + 1.07629\,(time)\right)^2$$

$$\widehat{Sales} = \left(10.26093 + 1.07629\,(time)\right)^2$$

STAT525 HOMEWORK#3 Solutions

1. NKNL Problem 2.57.

   a Testing $H_0 : \beta_1 = 5$ is really no different than testing $H_0 : \beta_1 = 0$ except that the reduced model is $E(Y) = \beta_0 + 5X$ instead of $E(Y) = \beta_0$. The full model has $n - 2$ error dfs while the reduced model has $n - 1$ error dfs.

   b The reduced model here is $E(Y) = 2 + 5X$. Here the the error dfs are n and the model degrees of freedom is 0.

2. NKNL Problem 2.61.

   Since $r_{XY} = r_{YX}$ and in simple linear regression $R^2 = r^2$, these values have to be equal. The hints suggest an alternative way to do this. The idea being that R^2 can be expressed as r^2 and then you can switch the variables.

$$R^2_{Y|X} = \frac{SSR_{Y|X}}{SSTO_Y} = \frac{(\sum (X_i - \overline{X})(Y_i - \overline{Y}))^2}{\sum (X_i - \overline{X})^2 \sum (Y_i - \overline{Y})^2} = \frac{SSR_{X|Y}}{SSTO_X} = R^2_{X|Y}.$$

   Or simply $R^2 = r^2$ and correlation coefficient r value is symmetric for X and Y.

3. NKNL Problem 3.19

   There are several ways to show this. Recall, however, that a positive relation between two variables means that these two variables have a positive correlation/covariance. Thus all methods should end up showing that there will often be a positive relationship between the residuals and the Y's.

   Consider computing the correlation between e_i and $\hat{Y}_i$. The numerator of this estimate is $\sum (\hat{Y}_i - \overline{Y})(e_i - \overline{e})$. Since $\overline{e} = 0$, this reduces to $\sum \hat{Y}_i e_i + \sum \overline{Y} e_i$. Since $\sum \hat{Y}_i e_i = 0$ and $\sum e_i = 0$ the numerator is zero and thus the correlation is zero.

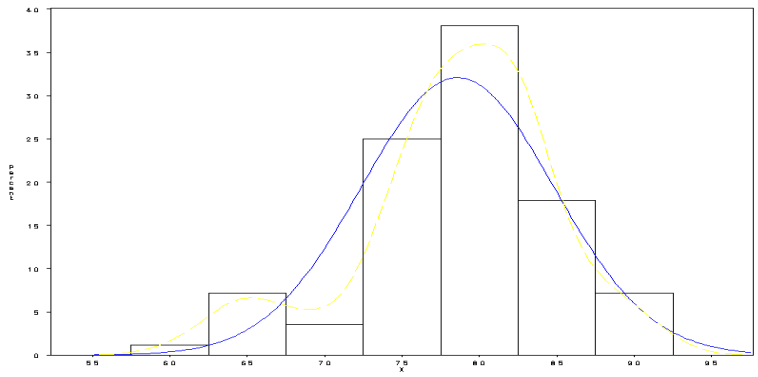
   We can do the same for the correlation between e_i and Y_i. The numerator of this estimate is $\sum (Y_i - \overline{Y})(e_i - \overline{e})$. Since $\overline{e} = 0$, this reduces to $\sum Y_i e_i + \sum \overline{Y} e_i$. Since $\sum e_i = 0$ the latter term is zero but the first term $\sum Y_i e_i = \sum (\hat{Y}_i + e_i) e_i =$SSE. Thus, this correlation will be positive.

4. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28.

   (a) Describe the distribution of the explanatory variable.

   Density is reasonably symmetric. The lower tail is a little unusual. The mean is 78.60 with a standard deviation of 6.22.

   (b) Run the linear regression to predict the county crime rate from the percentage of individuals having at least a high school diploma.



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93462942	93462942	16.83	<.0001
Error	82	455273165	5552112		
Corrected Total	83	548736108			

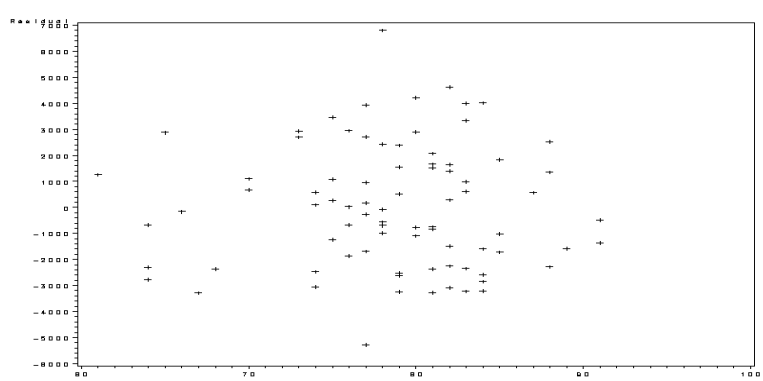
Root MSE	2356.29195	R-Square	0.1703
Dependent Mean	7111.20238	Adj R-Sq	0.1602
Coeff Var	33.13493		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	20518	3277.64269	6.26	<.0001
X	1	-170.57519	41.57433	-4.10	<.0001

(c) Plot the residuals versus the explanatory variable and briefly describe the plot noting any unusual patterns or points.

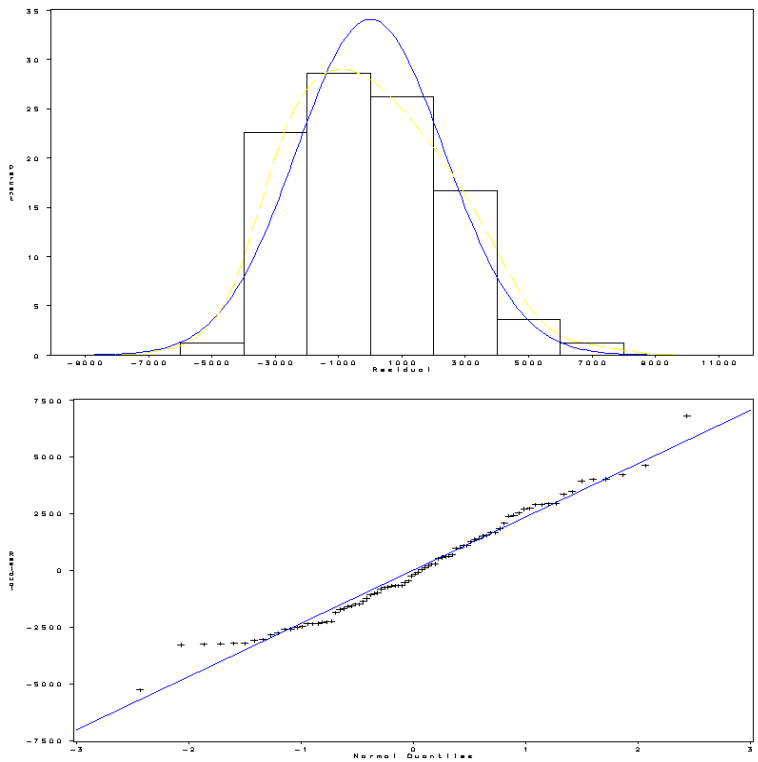
There does not appear to be anything unusual in the residual plot except that there are a couple of potential outliers (one large and one small)



(d) Examine the distribution of the residuals by getting a histogram and a normal probability plot of the residuals by using the `HISTOGRAM` and `QQPLOT` statements in `PROC UNIVARIATE`. What do you conclude?

The residuals appear slightly skewed compared to the normal distribution. Given the size of the data set, this isn't something I'd be too concerned about. If it were,

bootstrapping may be one way around the problem. The qqplot shows those potential outliers a bit better.



5. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28. Change the data set by changing the value of the crime rate for the last observation from 7582 to 758 (e.g., a typo). You can do this in a data step. For example,

```
DATA a2; SET a1; IF _n_ EQ 84 THEN y=758;
```

An alternative is to simply edit the data file.

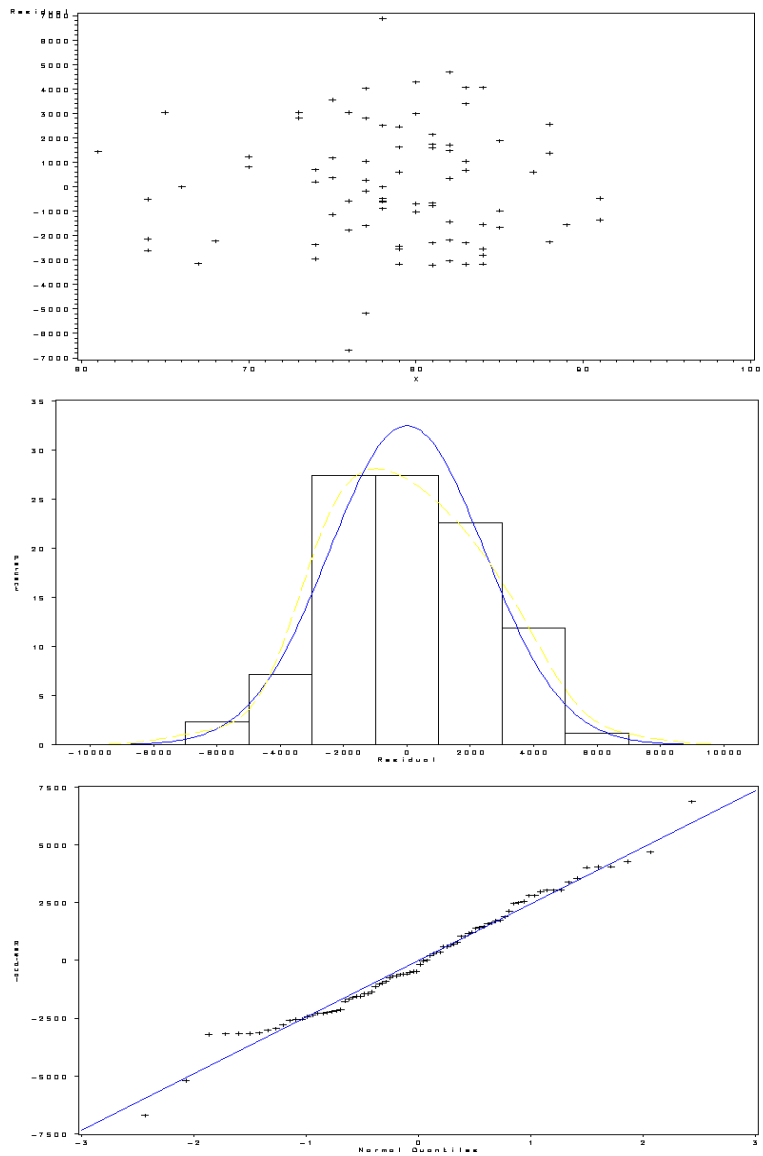
(a) Make a table comparing the results of this analysis with the results of the analysis of the original data. Include in the table the following: fitted equation, t-test for the slope, with standard error and P-value, R^2, and the estimate of σ^2. Briefly summarize the differences.

There is a substantial change in the error variance. This reduces the significance of Percentage explaining Rate and decreases R^2 but not substantially Because the typo is near the middle of the range of Percentage, the slope estimate does not change much. The intercept, however, is pulled down toward the outlier.

Data Set	Fitted Equation	$s(b_1)$	$H_0 : \beta_1 = 0$ P-value	R^2	MSE
Original	$\text{Rate} = 20518 - 170.58\text{Percentage}$	41.57	$< .0001$	0.1703	2356.29^2
Typo	$\text{Rate} = 20003 - 165.06\text{Percentage}$	43.60	0.0003	0.1488	2471.31^2

(b) Repeat parts (c) and (d) from problem 3.5 for this altered data set analysis and summarize how these plots help you to detect the unusual observation.

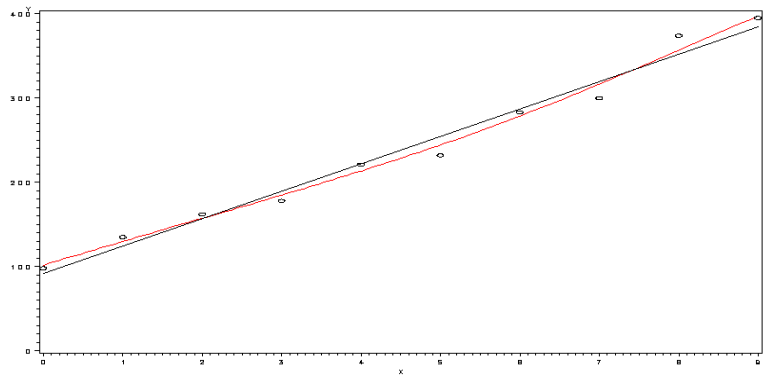
It is not that obivous these plots that there is a typo. Since the outlier is near the middle of the range of X, it does not affect the slope estimate.



6. (SAS Exercise) Use the **sales growth** data described in KNNL Problem 3.17.

(a) Generate a scatterplot of the data and discuss the appropriateness of using a linear regression model.

The scatterplot reveals a rather strong linear relationship. Perhaps the variance increase over X but the increase is small. You could probably get away with no transformation here.



(b) Using PROC TRANSREG, which power transformation of Y (i.e., value of λ) is most appropriate to use here?

The results suggest that a square root transformation is best here. The value $\lambda = 1$ is not in the 95% CI so it appears this is a helpful transformation.

Transformation Information

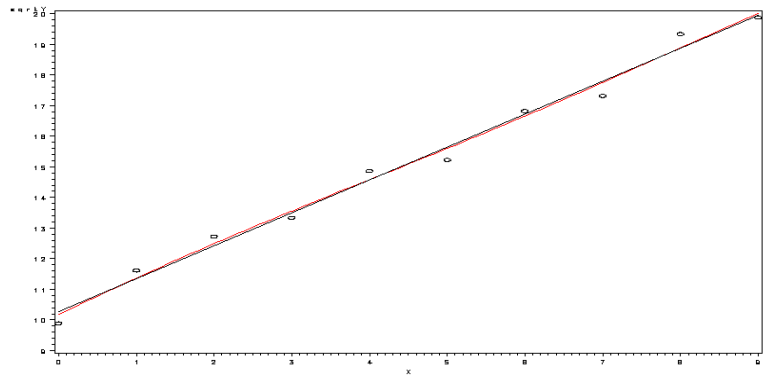
for BoxCox(Y)

Lambda	R-Square	Log Like	
-3.00	0.60	-50.9691	
-2.75	0.63	-49.3541	
-2.50	0.66	-47.7298	
-2.25	0.70	-46.0899	
-2.00	0.74	-44.4255	
-1.75	0.78	-42.7250	
-1.50	0.81	-40.9725	
-1.25	0.85	-39.1464	
-1.00	0.88	-37.2177	
-0.75	0.91	-35.1473	
-0.50	0.94	-32.8857	
-0.25	0.96	-30.3841	
0.00	0.98	-27.6568	
0.25	0.99	-25.0400	*
0.50 +	0.99	-23.7051	<
0.75	0.99	-24.7226	*
1.00	0.98	-27.0783	
1.25	0.97	-29.5670	
1.50	0.95	-31.8282	
1.75	0.94	-33.8405	
2.00	0.92	-35.6509	
2.25	0.90	-37.3078	
2.50	0.88	-38.8498	
2.75	0.85	-40.3059	
3.00	0.83	-41.6974	

< - Best Lambda
* - Confidence Interval
+ - Convenient Lambda

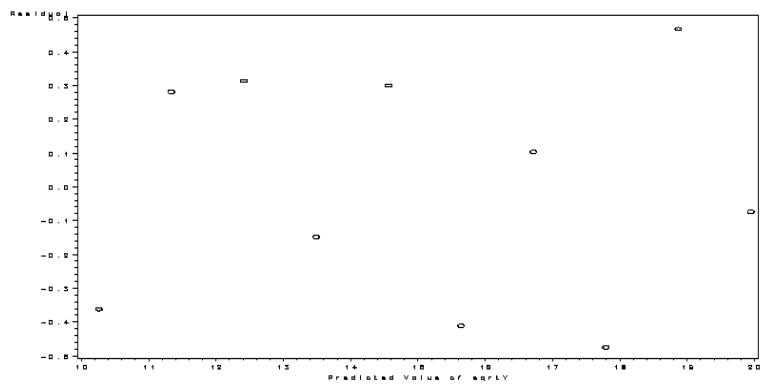
(c) Apply this transformation of Y and generate a scatterplot. Again comment on the appropriateness of using a linear regression model.

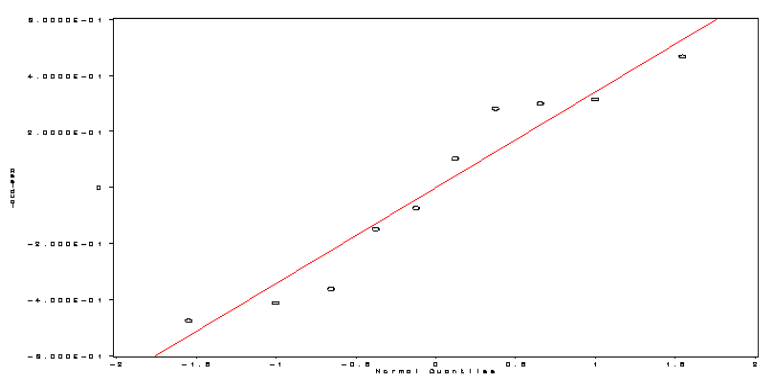
The square root transform does not appear to alter the linear relationship. The variability about the regression line seems more constant. This is also revealed in the residual plot below.



(d) Run the regression model using the transformed data and generate a residual plot (using X or $\hat{Y}$) and a normal probability plot. What do the plots show?

Residuals appear reasonable. No obvious violations.





(e) Express the estimated regression function in the original units.

The transformed model is $E(\sqrt{Y}) = 10.26 + 1.076X$. Transforming this back gives $E(Y) = 105.27 + 22.08X + 1.158X^2$

```
/*-------------------------------------------------------*/
/*                  SAS Codes for Q5                     */
data a1;
```

```
    infile 'D:\nobackup\tmp\CH01PR28.txt';
    input Y X;

proc univariate data=a1;
    var X;
    histogram X / kernel (L=2) normal;
run;

proc reg data=a1;
    model Y = X;
    output out=q5res r=resid;
run;

proc gplot data=q5res;
    plot resid*X;
run;

proc univariate data=q5res;
    qqplot resid / normal (L=1 mu=est sigma=est);
    histogram resid / kernel (L=2) normal;
run; quit;

/*-------------------------------------------------------*/
/*                 SAS Codes for Q6                      */
data a2;
    set a1;
    if _n_ eq 84 then Y=758;

proc reg data=a2;
    model Y = X;
    output out=q6res r=resid;
run;

proc gplot data=q6res;
    plot resid*X;
run;

proc univariate data=q6res;
    qqplot resid / normal (L=1 mu=est sigma=est);
    histogram resid / kernel (L=2) normal;
run; quit;

/*-------------------------------------------------------*/
/*                 SAS Codes for Q7                      */
data a3;
```

```
    infile 'D:\nobackup\tmp\CH03PR17.txt';
    input Y X;

symbol1 v=circle i=sm70 c=red;
symbol2 v=circle i=rl c=black;
proc gplot data=a3;
    plot Y*X=1 Y*X=2/overlay;
run;

proc transreg data=a3;
    model boxcox(Y) = identity(X);
run;

data a4;
    set a3;
    sqrtY = sqrt(Y);

proc gplot data=a4;
    plot sqrtY*X=1 sqrtY*X=2/overlay;
run;

proc reg data=a4;
    model sqrtY=X;
    output out=q7out p=pred r=resid;
run;

symbol v=circle i=none c=black;
proc gplot data=q7out;
    plot resid*pred;
run;

proc univariate data=q7out;
    qqplot resid / normal (L=1 mu=est sigma=est);
run; quit;
```